

Research Paper: Multi-Agent AI Systems for Drug Repurposing

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Abstract

Drug discovery remains slow, risky, and expensive. Industry analyses estimate the overall probability of a molecule entering Phase I ultimately achieving approval close to 10%, with company-level averages typically ranging between 8 and 23%, making drug repositioning an attractive and less risky path to new therapies. This capstone project builds a staged, evidence-based pipeline from the public Kaggle Drug Repositioning dataset to generate and prioritize repositioning hypotheses, then selectively consults advanced systems as external auditors. We start by engineering transparent features such as target promiscuity, pathway overlap via DrugBank, literature co-mentions via PubMed, and lightweight chemoinformatics descriptors, and then train tabular baselines alongside a simple deep model. On the retained data, the deep model achieved a test MSE of 135.968 and an MAE of 9.002, establishing a measurable but improvable signal justifying the use of structure-aware controls. We then propose to audit the top-ranked pairs using a graph-based model (TxGNN) to inject a global biomedical graph context and a lightweight agentic loop, modeled after recent multi-agent frameworks such as PharmAgents, LLM Agent Swarm, and Google’s AI co-scientist, to extract mechanisms, synthesize the literature, and suggest upward or downward adjustments with plain-language justifications. The expected contribution is a reproducible and auditable model combining the traceability of ML based on feature engineering on open data with the breadth and depth of reasoning of modern graphical and multi-agent systems, producing therapeutic and pathological hypotheses ranked and supported by evidence, suitable for triage by experts and future laboratory validation.

Getting a new treatment from the bench to the bedside remains extremely challenging: over the period 2011–2020, the probability of Phase I approval was only about 7.9%, with high attrition at each phase, which partly explains the decade-long delays and costs in the billions. These macro constraints make drug repurposing, the search for new indications for already approved or shelved compounds, particularly attractive because safety profiles and manufacturing pathways are widely known, but the evidence needed to link drugs to new diseases is scattered across a huge biomedical graph that no human can decipher from end to end. To tame this evidence, the field has evolved beyond single-task models to graph-based models and agentic (multi-agent) AI. A prominent example is TxGNN, introduced in Nature Medicine, which performs sampling-free reassignment by reasoning over a clinician-centric knowledge graph. It achieves significant gains over previous benchmarks and can evaluate candidates for approximately 17,000 diseases, including rare conditions with sparse data. Meanwhile, multi-agent systems coordinate specialized AI “colleagues” who generate, debate, and refine hypotheses: Google’s AI Co-Scientist, based on Gemini 2.0, operationalizes a generate-debate-evolve loop, while research frameworks such as PharmAgents and LLM Agent Swarm (PharmaSwarm) simulate a virtual pharmaceutical industry of LLM-driven agents, spanning target discovery, literature search, mechanistic assessment, and safety screening. Together, these axes suggest more performant and explainable hypothesis generation than monolithic models. This graduation project adopts a progressive, evidence-driven strategy: we start with a transparent machine learning database based on public data, then use sophisticated graph and multi-agent systems as auditors to confirm, refute, or adjust these results. Concretely, we anchor three CSV files linked to the Kaggle Drug Repositioning dataset: drugsInfo, diseaseInfo, mapping, we design interpretable features such as target promiscuity, pathway

overlap via DrugBank, literature co-mentions via PubMed, and we train a simple deep model and tabular learners to produce an initial list of ranked hypotheses. This public and licensed corpus is ideal for an academic prototype and can be reproducibly enriched with selected resources.

Baseline signal and its limitations. Our first pass, a very simple deep learning model trained on technical features, achieved a test MSE of 135.968 and an MAE of 9.002 on the holdout data. Simply interpreted, the predictions deviate from the ground truth by about nine units (on the chosen target scale) on average. These results are sufficient to demonstrate signal learning in Kaggle features, but imperfect enough to warrant escalation: some relationships remain invisible to shallow features or sparse literature inventories and likely require a global graph structure and deeper reasoning to resolve. Therefore, we will treat the first N elements of the baseline as preliminary hypotheses, plausible but not yet convincing candidates. These results are from our experience; we will discuss below how we plan to improve them. Next, we selectively scale up advanced systems as independent, structure-aware reviewers. We query TxGNN scores, when available, to verify whether a global biomedical graph validates or disputes specific pairs. We route disagreements to a lightweight agentic audit, a scripted generation, retrieval, and critique loop that extracts evidence and literature snippets and then offers upward or downward adjustments to the baseline ranking, accompanied by plain-language justifications. This model mirrors Google's reporting for its co-scientists and what open research prototypes such as PharmAgents, LLM Agent Swarm, advocate: letting specialist colleagues debate and develop ideas, while maintaining a transparent baseline so humans can examine assumptions, uncertainties, and provenance.

Computational drug repurposing began with similarity heuristics and single-task machine learning models trained on carefully designed features. Recent studies demonstrate a decisive shift toward deep architectures, capable of natively handling heterogeneous and high-dimensional biomedical data and generalizing other indications. These analyses highlight the benefits of integrating chemical, biological, and clinical signals, but also highlight recurring challenges, data incompleteness, publication bias, and the need for transparent and reproducible pipelines to gain clinician trust. The rise of graph base models, which reason on drug-protein-disease knowledge graphs, is a major advance. TxGNN, presented in Nature Medicine, is a prominent example: trained on a large clinician-centric medical knowledge graph, it performs flawless repurposing by classifying indications for 17,080 diseases, including rare conditions. Published results show substantial gains over previous baselines for indication and contraindication tasks, as well as utilities for clinician inspection. The TxGNN project's public page details the improvements, up to approximately 49% indication accuracy, and offers a human-AI explorer to visualize rationales. These properties make graph base models attractive second-opinion engines for auditing or supplementing simpler machine-learning tabular benchmarks.

Alongside advances in representation learning, the field is moving from monolithic models to multi-agent AI systems in which specialized agents collaborate, critique, and iteratively refine hypotheses, an organizational model that mirrors that of real-world research teams. PharmAgents proposes a virtual pharmaceutical industry composed of more than 20 LLM-driven agents that cover the entire pipeline, from target discovery to in silico toxicity assessment, integrating external tools and domain-specific models. The authors report improved lead quality through structured knowledge exchange and self-improvement across cycles.

Similarly, PharmaSwarm, an LLM Agent Swarm, presents hypothesis-driven discovery as an orchestrated exchange between specialized agents, with a central reviewer continuously ranking proposals based on plausibility, novelty, and safety, while maintaining shared memory for self-evolution. This modular design addresses the limitations of individual LLMs, such as lack of iterative memory and limited tool specialization, and integrates seamlessly into regulated scientific workflows. There are also domain-specific agentic systems for repurposing, such as DrugAgent, which combines LLM-based reasoning, knowledge graph search, and model-based scoring to produce interpretable suggestions for drugs and diseases. DrugAgent's emphasis on explanation, such as pathway evidence and textual justifications, directly highlights barriers to clinical adoption. Taking the paradigm further, Google's AI co-scientist uses Gemini-based multi-agent loops, generating then debating then evolving, to propose scientifically novel hypotheses, with case studies spanning drug repurposing, target discovery, and antimicrobial resistance. Public documents describe the architecture and report prospective validations such as epigenetic targets for liver fibrosis and antitumor activity in cellular models of AML, illustrating how agentic orchestration can increase novelty and yield. Industry reports corroborate these demonstrations and position the co-scientist as a model of collaborative, hypothesis-driven AI. In these systems, two cross-cutting themes recur: evidence integration and explainability. Graph models like TxGNN integrate multi-source structures, such as drug-target-disease relationships, and expose interpretable artifacts, such as graph pattern saliency, facilitating clinical and error analysis. Agentic frameworks go further by associating model scores with literature snippets, pathway overlap, and uncertainty estimates, results closer to domain experts' expectations for translational decision-making. Analyses of AI in drug discovery highlight the need for provenance tracking, model transparency, and calibrated trust to mitigate automation bias; agent

teams with explicit explanatory roles are a pragmatic response. Crucially, explainability is about more than presentation; it can fuel closed-loop improvement. For example, when a reference tabular model proposes a drug-disease correspondence primarily due to target promiscuity, an agentic audit can cross-reference pathway specificity and literature support, thereby downgrading candidates lacking mechanistic alignment and favoring those with converging data. This two-tiered, human-centric workflow, with first a transparent baseline and then sophisticated agent/graph audits, is increasingly advocated to balance speed and rigor.

Methodological advances depend on accessible and verifiable data. The Kaggle Drug Repositioning dataset is a widely used public corpus containing `drugsInfo.csv`, `diseaseInfo.csv`, and `mapping.csv` tables such as SMILES, target lists, therapeutic categories, and known indication pairs. While smaller and more complex than proprietary resources, it enables reproducible method development, rapid iterations, and fair comparisons, especially when complemented by authoritative knowledge bases such as DrugBank, UniProt, and literature signals. Because agentic systems often rely on tool retrieval and use, public datasets like Kaggle's provide realistic supports for comprehensive demonstrations, free from legal constraints. That said, open corpora also crystallize data risks that sophisticated models can mask: categorical imbalance (dominance of oncology and anti-infectives), incomplete target annotations, and heterogeneous naming/synonymy across sources. The literature consistently advocates standardized cleaning, harmonization via canonical identifiers, and robust documentation, including through DVC-monitored pipelines, to ensure that improvements are methodological and not the result of ad hoc preprocessing. In the context of agent-based AI, explicit provenance and memory structures allow for the recording of who (which agent) did what (which transformation) with which source, thus promoting reproducibility and auditability.

Overall, my work suggests a pragmatic research strategy: start with a transparent baseline based on public data and then move to advanced systems serving as auditors and amplifiers. First, a tabular or tree-based machine learning model, equipped with domain-specific features such as target promiscuity, disease-gene overlap, and co-mentions in the literature, establishes a reproducible and interpretable baseline. Then, graph-based models like TxGNN can verify whether the overall structure confirms or refute these predictions, especially for rare diseases where tabular features are weak. Finally, agent-based audits retrieve targeted evidence, synthesize mechanistic rationales, and propose adjustments to the baseline ranking, such as reweighting by pathway specificity or contraindication risk, thus producing a list that is both data-driven and argument-rich. This phased approach builds on the complementarity documented in the literature: tabular machine learning for speed and clarity; graphs for long-term biological context. Agents enable iterative reasoning and explanation, while controlling costs and ensuring reasoning traceability. Despite evident momentum, important gaps remain, which this final project can help highlight. Prospective validation is still scarce: many agent and graph systems report retrospective measurements, but fewer demonstrate confirmed laboratory results on novel and out-of-distribution hypotheses. Calibration and uncertainty also require systematic attention, especially when aggregating signals from heterogeneous agents. Finally, governance and safeguards, ensuring the reliability of recovery sources, the traceability of claims, and the early detection of failure modes, will determine the transition of agentic AI from convincing demonstrations to reliable collaborators. The phased design proposed here is well positioned to explore these questions on open data, while remaining extensible to richer corpora and experimental partnerships.

This project is based on the publicly available Kaggle drug repositioning dataset (Aria Hassanali Aragh, n.d.). This curated resource contains 1,410 drug entries, including their names, chemical structures, expressed via SMILES strings, known target proteins, associated therapeutic classes, and documented indications. This dataset was chosen for its accessibility, comprehensiveness, and relevance for identifying new uses for existing drugs. Although not exhaustive, it provides a valuable tool for hypothesis generation and a starting point for building interpretable models that can facilitate drug repositioning tasks. Each row in the dataset represents a unique drug, providing a solid foundation for feature engineering and training machine learning models. However, relying solely on the Kaggle dataset is insufficient due to several inherent limitations: it prioritizes well-studied compounds, lacks up-to-date bioactivity or side effect profiles, and offers limited mechanistic depth. To address these shortcomings, three key external knowledge sources will be integrated to enrich the dataset and support downstream analysis. First, DrugBank will be used to extract detailed drug-drug interactions, molecular mechanisms of action, and ATC codes to identify functional classes (Wishart et al., 2018). Second, UniProt (UniProt Consortium, 2023) will be used to enrich protein target information by mapping drug targets to protein families, known biological functions, and their involvement in disease pathways. This will enable better interpretation and construction of meaningful biological features. Third, literature-derived features will be generated by mining PubMed resumes using electronic utilities to calculate co-mention frequencies between drug and disease entities. These counts will serve as proxies for implicit evidence or historical attention in the biomedical literature, highlighting overlooked candidates or false positives. Integration of these sources will allow the construction of composite features for each drug, including: number of targets and pathway overlaps, co-mention scores in disease-specific literature, structural

similarity to other drugs repurposed using chemoinformatics tools such as RDKit (Landrum, 2023), and drug promiscuity measures to assess the likelihood of a drug interacting with off-target pathways. The enriched dataset will be used both to create interpretable base models and to power more advanced reasoning systems later in the pipeline. Feature harmonization and normalization processes will ensure that variables from different sources are properly matched to the core dataset and usable in tabular and graphical machine learning architectures. This combined approach aims to ensure that the resulting models are not only accurate but also based on diverse and biologically informative data. By leveraging a freely available dataset and enriching it exclusively with reliable biomedical sources, the project ensures transparency, reproducibility, and scientific rigor, thus meeting the needs of academic research and practical drug development processes.

To explore the viability of machine learning for drug repurposing, this project adopts a two-stage methodological framework. The first stage involves traditional machine learning pipelines with engineered features derived from structured drug data. The second stage introduces multi-agent AI systems and large language model-based reasoning agents, which analyze, evaluate, and potentially enhance or correct the outputs from the initial model. This hybrid approach bridges interpretable modeling and advanced AI collaboration to deliver both explainability and depth. The process begins with extensive feature engineering using the base Kaggle Drug Repositioning Dataset and its augmented sources. Several categories of features are constructed. Chemical features include molecular weight, number of hydrogen bond donors/acceptors, and topological polar surface area, derived using cheminformatics libraries like RDKit (Landrum, 2023). These features serve as structural fingerprints of each compound.

Biological features are extracted from DrugBank and UniProt, such as the number of known targets, target pathway involvement, and protein-protein interaction degree for each target gene. These provide a pharmacological context to each drug’s mechanism of action. Text-derived features are created by querying PubMed using E-utilities to compute the frequency of co-mentions between a drug and a disease class. These scores are normalized using TF-IDF-style transformations to capture the informativeness of co-occurrences without biasing toward common diseases. Once the features are extracted and cleaned, the dataset is split into training and testing sets. A baseline deep learning model is implemented using TensorFlow on Google Colab. The model is a fully connected feedforward neural network that predicts therapeutic match scores, proxy labels for potential new indications, based on input features. The dataset’s numerical values are normalized, and categorical values are one-hot encoded or embedded where necessary. During initial experiments, the model yielded a test Mean Squared Error (MSE) of 135.97 and a Mean Absolute Error (MAE) of 9.00, suggesting moderate predictive power with room for improvement. Recognizing the limits of this initial model, the second stage leverages insights from recent advancements in multi-agent systems for scientific hypothesis generation. Inspired by frameworks such as PharmaSwarm (Song et al., 2025) and PharmAgents (Gao et al., 2025), a simplified LLM-driven architecture is introduced in which autonomous agents reflect, debate, and refine hypothesis outputs from the earlier model. Each agent is assigned a domain-specific task, such as mechanism-of-action verification, literature triangulation, or toxicity flagging. These agents use results from the first model as starting points and apply additional reasoning layers through prompt-engineered queries, embedding lookups, or symbolic rule checking. This agent-based model does not replace the neural model but acts as a post-hoc analyst, aimed at refining, prioritizing, or filtering its outputs. For example, if a drug-disease pair

is highly ranked by the neural model but has sparse literature evidence, the agent may flag it for further study or downgrade its recommendation score. In cases where the model misses a known target due to feature sparsity, an LLM agent may surface supporting literature, effectively boosting recall. By combining interpretable modeling with agentic refinement, the methods employed in this project not only demonstrate predictive capacity but also align with emerging paradigms in AI-assisted scientific discovery (Yang et al., 2025; Google Research, 2025). The result is a dual-layered architecture capable of handling both tabular and unstructured data, anchored in replicability, transparency, and continuous improvement.

The evaluation framework for this project is designed to reflect the dual-layered methodology involving both a baseline machine learning model and an AI-driven multi-agent refinement stage. The goal is to assess not only the predictive power of the initial deep learning model but also the added value of agentic post-processing in improving interpretability, relevance, and novelty of the drug-disease hypotheses generated. The analysis begins with standard model performance metrics applied to the baseline neural network trained on engineered features. Mean Squared Error (MSE) and Mean Absolute Error (MAE) are used to assess regression performance, given the continuous output of repurposing likelihood scores. In the pilot experiment, a test MSE of 135.97 and an MAE of 9.00 were observed, indicating that the model was able to detect patterns in the drug data but also revealed significant residual errors, highlighting room for optimization. In subsequent iterations, improvements such as hyperparameter tuning, model architecture adjustments, or advanced regularization techniques will be considered to reduce these errors. Beyond raw metrics, qualitative evaluation is introduced through human-in-the-loop validation. The top 10 predictions from the model will be

reviewed for biological plausibility by cross-referencing external resources like PubMed, DrugBank, and clinical trial databases. Additionally, literature co-mention analysis and mechanistic validation such as checking if the drug targets are implicated in the disease pathway will be carried out by agents and domain experts. This ensures that the results are not just mathematically accurate, but also biologically meaningful—an essential factor in translational medicine (Wan et al., 2025). To evaluate the impact of the agent-based enhancement layer, a comparative analysis will be conducted. Hypotheses generated solely by the machine learning model will be compared with those refined or modified by the multi-agent system. Evaluation criteria will include relevance, novelty, and explainability. Metrics such as precision@k, human-rated plausibility scores, and coverage of high-impact literature evidence will be used to assess added value. This layered analysis strategy provides a balanced view of both algorithmic accuracy and scientific credibility (Gao et al., 2025; Song et al., 2025). However, despite the promise of this approach, several limitations must be acknowledged. First, the Kaggle Drug Repositioning Dataset, while rich in scope, is inherently biased toward well-studied drugs and indications. Rare diseases and less-published compounds may be underrepresented, leading to poor generalization in such cases. Second, the reliance on co-mention frequencies from literature as a proxy for drug-disease relationships introduces noise, as high co-occurrence does not always imply therapeutic relevance. Third, the deep learning model, despite feature engineering, operates as a black box, and its predictions may lack transparency unless augmented with explainable AI tools such as SHAP (Lundberg & Lee, 2017). Additionally, agentic systems, while powerful, bring their own challenges. The performance of LLM agents can be inconsistent depending on prompt design, context windows, or the availability of relevant background information. There is also a computational overhead involved in orchestrating multiple agents

and memory-sharing mechanisms, which could be impractical for deployment in resource-constrained environments. Moreover, because these systems are built on large language models that are trained on static data, they may hallucinate or generate confident but incorrect explanations, posing risks to scientific integrity if not carefully monitored (Google Research, 2025; Inoue et al., 2024).

The integration of artificial intelligence into biomedical discovery, particularly in drug repurposing, demands a thoughtful consideration of ethical principles across multiple dimensions. While the project primarily leverages publicly available data and computational models, the downstream implications of its outcomes touch on human health, equity, transparency, and trust in automated scientific systems. Ethical diligence is therefore not a peripheral concern, but central to the design, development, and deployment of this project. One of the foremost ethical concerns is algorithmic bias. The Kaggle Drug Repositioning Dataset, as well as supplemental sources such as DrugBank and literature databases, inherently favor well-studied drugs and diseases. As a result, the machine learning model may exhibit a bias toward common, well-funded diseases such as cancer or cardiovascular conditions, while neglecting rare or neglected tropical diseases that disproportionately affect underrepresented populations. This asymmetry could inadvertently perpetuate existing health inequities if not addressed through dataset balancing, algorithmic adjustments, or targeted filtering mechanisms (Nweke et al., 2025). Another key ethical issue is explainability and interpretability, particularly given that both machine learning and multi-agent language model systems can behave like black boxes. Without mechanisms for transparent reasoning, such as SHAP values (Lundberg & Lee, 2017) or agent-generated plain-language justifications (Song et al., 2025), stakeholders, including clinicians,

patients, and regulatory bodies, may be unable to trust or verify the system's recommendations. This is especially critical in healthcare contexts, where opaque AI-driven decisions may have real-world consequences. Thus, this project emphasizes the inclusion of interpretable outputs at every stage, from feature-engineered ML results to multi-agent refinements. Data privacy is a lesser concern in this case, as the project intentionally avoids patient-level data and relies exclusively on aggregated, de-identified, or open-access datasets. However, as the system matures and if future iterations incorporate real-world clinical records (as in Chen et al., 2021), it will be essential to ensure compliance with data protection regulations such as HIPAA in the U.S. or GDPR in Europe. At that point, implementing rigorous data governance, anonymization, and auditability protocols will be necessary to safeguard sensitive health information. Furthermore, the use of LLM-based agents introduces the potential for AI hallucinations, confident but inaccurate claims not grounded in evidence, which can mislead researchers or distort scientific reasoning if not properly monitored. This project mitigates such risks by including critical validation layers, such as co-mention frequency analysis, cross-checking with curated biomedical knowledge bases such as UniProt, SIDER, and manual review when possible. Still, as agentic systems gain more autonomy, the ethical challenge of controlling their behavior, and ensuring their outputs align with truthfulness and scientific rigor, becomes paramount (Google Research, 2025; Inoue et al., 2024). Lastly, responsible innovation and equitable access must be considered. The vision of this project is to democratize drug repurposing by offering a reproducible, transparent, and open-source framework. Unlike proprietary pipelines developed by large pharmaceutical companies, this work aims to empower academic labs, NGOs, and low-resource institutions with intelligent tools for hypothesis generation. However, care must be taken to

document the system thoroughly, avoid over-promising results, and communicate limitations clearly to avoid misuse or unrealistic expectations.

To conclude, the initial deep learning experiment on the Kaggle Drug Repositioning dataset generated an MSE of 135.968 and an MAE of 9.002, providing informative benchmarks for the project's fundamental modeling. While these error metrics reflect moderate predictive performance, indicating average amplitude deviations of approximately 9 between predicted and actual values, they are reasonable considering the dataset's heterogeneity and limited feature complexity. Notably, in related fields such as regression tasks in medicinal chemistry, MAE results in this range are considered promising when interpretable chemical descriptors or simple architectures are used. These results support the role of a transparent, feature-optimized baseline: it offers a reproducible starting point with quantifiable error and interpretability. More importantly, the baseline's moderately high performance suggests that it captures significant signal in the data, target promiscuity, therapeutic breadth, and co-mentions in the literature, while remaining imperfect enough to benefit from additional layers of reasoning. Herein lies the value of the proposed multi-agent augmentation: systems like TxGNN (graph foundation model) and LLM-based audit agents can analyze top benchmark results and propose subtle adjustments, favoring promising pairs with strong structural or mechanistic support and demoting likely artifacts lacking external validation. Rather than discarding the benchmark, these advanced agents act as domain-aware reviewers, enriching predictions with complementary context and biological information. In essence, the simple deep learning model, with its test MAE around 9, serves as a baseline, not a final product. The moderate error leaves room for improvement by

integrating graphical context and agentic debate. By layering these capabilities on top of a transparent and explainable base, the project builds a pipeline that is both reproducible and scientifically grounded, delivering prioritized, human-readable drug-disease hypotheses that are stronger, better justified, and better suited to downstream experimental validation.

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